

Assignment 5

Ethan

Set Up

Packages

```
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

#Wrap Axis Label
library(scales)

#reading .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# visualizing
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(
```

```

  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects"
)

# site information
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+")
)

```

Necessary Code from Class

```

# creating new object from sites
ncos_sites <- sites %>%
  # filtering to only include sites at NCOS, sampled quarterly
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

# creating new object from taxon_list
taxon_list_clean <- taxon_list %>%
  # converted taxon names to snake case
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))

water_quality_clean <- water_quality %>%
  clean_names() %>%
  semi_join(ncos_sites, by = "site") %>%
  unite("date_site", date, site, remove = TRUE) %>%
  group_by(date_site) %>%

```

```

summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) %>%
ungroup() %>%
filter(date_site != "2022-08-12NVBR")

aq_ins_clean <- aq_ins %>%
  clean_names() %>%
  semi_join(ncos_sites, by = "site") %>%
  rename(date = date_on_vial) %>%
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         annelida_polychaete,
         diptera_chironomid) %>%
  filter(sample_type %in% c("FB250", "FB 250")) %>%
  pivot_longer(
    # columns to make longer
    cols = ostracod:annelida_polychaete,
    # new column for the old column names
    names_to = "taxon",
    # new column for the values in each of the old columns
    values_to = "abundance"
  ) %>%
  group_by(date, site, taxon) %>%
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) %>%
  ungroup() %>%
  unite("date_site", date, site, remove = FALSE) %>%
  left_join(water_quality_clean, by = "date_site") %>%
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) %>%
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "West Pond",

```

```

    site == "NDV" ~ "Devereux Creek")) %>%
mutate(site_full = fct_reorder(site_full,
                               med_sal,
                               .fun = "median",
                               .na_rm = TRUE),
       site_full = fct_rev(site_full))

copepods <- aq_ins_clean %>%
  filter(taxon == "copepod") %>%
  mutate(log_ave_lit = log(ave_lit + 1))

```

1. Visualizing differences across sites in major taxon

a. Planning the Figure

- x-axis: Salinity (ppt)
- y-axis: taxon
- geometry to represent average abundance/L: `geom_jitter()`
- geometry to represent medians: `stat_summary()`

b. Wrangling the data

```

ostracods <- aq_ins_clean %>%
  filter(taxon == "ostracod") %>%
  mutate(log_ave_lit = log(ave_lit + 1))

slice_sample(ostracods, n = 10)

```

A tibble: 10 x 24

	date_site	date	site	taxon	ave_lit	med_ph	med_sal	med_do
	<chr>	<dtm>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	2023-03-03_NPB2	2023-03-03 00:00:00	NPB2	ostr~	0.133	7.71	1.98	9.59
2	2022-05-06_NMC	2022-05-06 00:00:00	NMC	ostr~	0	9.96	1.45	16.5
3	2022-11-19_NPB1	2022-11-19 00:00:00	NPB1	ostr~	2	7.26	2.47	3.96
4	2023-09-15_NMC	2023-09-15 00:00:00	NMC	ostr~	0.4	NA	NA	14.2
5	2023-02-12_NEC	2023-02-12 00:00:00	NEC	ostr~	2.13	NA	NA	NA
6	2023-06-06_NPB1	2023-06-06 00:00:00	NPB1	ostr~	0.533	7.48	NA	2.38

```

7 2023-05-04_NEC 2023-05-04 00:00:00 NEC ostr~ 0.667 8.05 NA 8.55
8 2023-11-12_NMC 2023-11-12 00:00:00 NMC ostr~ 0.533 NA 47 14.0
9 2022-11-18_NVBR 2022-11-18 00:00:00 NVBR ostr~ 0 NA NA NA
10 2021-11-12_NPB 2021-11-12 00:00:00 NPB ostr~ 79.2 NA NA NA
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
# Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
# Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
# comments <chr>, site_full <fct>, log_ave_lit <dbl>

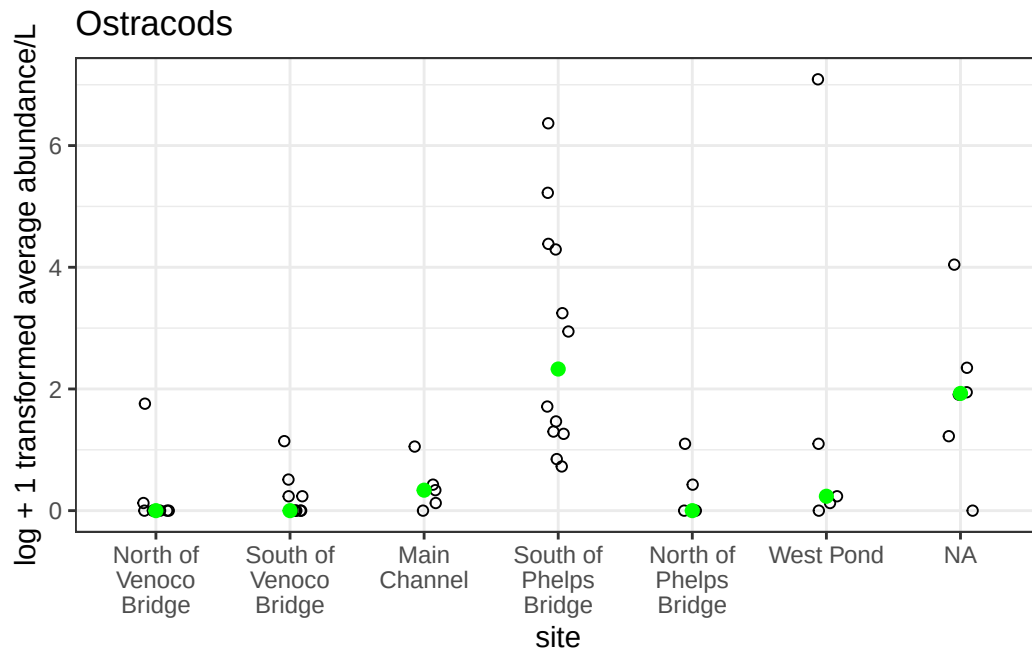
```

c. Coding the figure

```

ostracod_plot <- ggplot(data = ostracods,
                        mapping = aes(x = site_full,
                                      y = log_ave_lit))+
  geom_jitter(height = 0,
             width = 0.1,
             shape = 21) +
  stat_summary(geom = "point",
             fun = median,
             color = "green",
             size = 2)+
  scale_x_discrete(labels = label_wrap(width = 10)) +
  labs(x = "site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracods") +
  theme_bw()
ostracod_plot

```



d. Creating a multipanel figure

```
# creating an object of a salinity plot
salinity_plot <- ggplot(data = aq_ins_clean,
                        mapping = aes(x = site_full,
                                      y = med_sal)) +

  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  stat_summary(geom = "point",
              fun = median,
              color = 'red') +
  scale_x_discrete(labels = label_wrap(width = 10)) +
  labs(x = "site",
       y = "Median Salinity (ppt)",
       title = "Site Salinity")

# creating a plotting object of copepod abundance
copepod_plot <- ggplot(data = copepods,
```

```

mapping = aes(x = site_full,
              y = log_ave_lit))+
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +
stat_summary(geom = "point",
            fun = median,
            color = "blue",
            size = 2)+
scale_x_discrete(labels = label_wrap(width = 10)) +
labs(x = "site",
     y = "log + 1 transformed average abundance/L",
     title = "Copepod") +
theme_bw()

```

```

salinity_plot /
(copepod_plot | ostracod_plot)

```

